

# An Explainable Machine Learning CDSS with Clinical Baseline Imputation and Youden-Thresholding for Bedside ICU Mortality Prediction

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**Abstract**—Early identification of patient deterioration in Intensive Care Units (ICUs) is critical to improving survival rates, yet clinical environments suffer from alarm fatigue due to high false-positive warning rates. This study presents the design of an end-to-end, leak-free Explainable AI (XAI) bedside Clinical Decision Support System (CDSS) for in-hospital mortality risk prediction using physiological data from the initial 24 hours of an ICU stay. We evaluated seven machine learning classifiers and two ensembles on the MIMIC-IV Clinical Demo dataset (100 patients, 140 stays, 11% mortality rate). To regularize the feature space without over-relying on synthetic samples, we designed a clinical preprocessing pipeline featuring temperature unification, outlier scrubbing, and clinical baseline imputation (imputing missing laboratory draws with healthy reference values to simulate medical normalcy). To address severe class imbalance, we applied out-of-fold Youden's J threshold optimization. An ExtraTrees Classifier emerged as the champion model, utilizing a fold-averaged decision threshold of 0.39. On the holdout test set, the champion model achieved an accuracy of 0.5500, a sensitivity of 0.5000, and a specificity of 0.5556 (ROC-AUC = 0.5556, PR-AUC = 0.1603). Transparency is achieved via a Tree SHAP engine, which maps model decisions to local bedside biophysical risk drivers. The system is deployed as an interactive Streamlit application with rule-based clinical guides and automated Patient Chart PDF generation to support clinical decision-making.

**Keywords**—ICU Mortality Prediction, Explainable AI, Clinical Decision Support, ExtraTrees Classifier, Youden's J Threshold, Tree SHAP, MIMIC-IV Clinical Demo.

## I. INTRODUCTION

The Intensive Care Unit (ICU) is a high-acuity environment where patients require continuous monitoring and rapid, life-critical interventions. Critical care teams are inundated with massive, heterogeneous data streams generated by bedside telemetry monitors, ventilators, and frequent laboratory assessments. This data density frequently results in cognitive overload and alarm fatigue, where clinical staff desensitize to warning sounds, occasionally missing critical physiological deterioration cues.

Machine learning (ML) models trained on Electronic Health Records (EHR) hold the potential to act as early-warning systems, stratifying patients by mortality risk. However, standard clinical ML models often suffer from two major flaws: (1) data leakage, resulting from features extracted immediately before death or discharge, rendering the model useless for early intervention; and (2) the black-box nature of complex ensembles, which limits clinician bedside trust.

To address these limitations, we present a feasibility study for an Explainable AI-based Clinical Decision Support System (CDSS). The model strictly restricts its feature extraction to the first 24 hours of a patient's first ICU admission, establishing a leak-free observation window. We demonstrate the system's viability using the publicly available MIMIC-IV Clinical Demo dataset.

## II. RELATED WORK

ICU risk scoring systems, such as the Simplified Acute Physiology Score (SAPS II) and the Acute Physiology and Chronic Health Evaluation (APACHE), have long been used to estimate mortality risk. While robust, these scores are calculated statically at the end of the first 24 hours, failing to capture continuous temporal trajectories.

With the growth of EHR databases like MIMIC, researchers have deployed machine learning classifiers (e.g., Logistic Regression, Random Forest, Gradient Boosted Trees) to predict outcomes. Despite high statistical accuracy, clinical adoption remains low. Clinicians require interpretability—knowing why a model flags a patient. Recent advancements in Explainable AI (XAI), particularly SHAP (SHapley Additive exPlanations) based on

cooperative game theory, offer a mathematically consistent method to decompose individual patient predictions into biophysical risk contributions.

### III. PROBLEM STATEMENT

Given a patient's demographics, continuous bedside vital signs, and laboratory panels collected during the first 24 hours of their first ICU stay, the objective is to predict the probability of in-hospital mortality (hospital\_expire\_flag).

The prediction must be made under two main constraints: (1) Clinical Actionability: The model must operate on early observations to allow preventive clinical interventions. (2) Imbalance Resilience: The model must handle severe class imbalance (11.0% mortality rate) without collapsing into predicting survival for all cases (majority-class collapse).

### IV. PROPOSED METHODOLOGY

The proposed CDSS is structured into three primary layers: (1) Clinical Data Ingestion & Preprocessing: Parses raw EHR tables, isolates the target cohort, converts temperature scales, and scrubs sensor noise. (2) Predictive Machine Learning Engine: Compiles 6 statistical aggregates over 24 hours, scales features, and trains a regularized ExtraTrees Classifier utilizing Youden's J threshold optimization. (3) Actionable Clinical Interface: Computes SHAP values, renders risk alerts, and generates bedside PDF charts.

### V. DATASET DESCRIPTION & CLASS DISTRIBUTION

Experiments were performed using the publicly available MIMIC-IV Clinical Demo Dataset (100 ICU patients), which contains a de-identified sample of EHR records (version 2.2). The cohort tracks 100 unique adult patients across 140 ICU stays. The outcome target hospital\_expire\_flag exhibits a severe class imbalance, with 15 stays resulting in in-hospital mortality (11.0%) and 125 stays resulting in survival (89.0%).

The data was partitioned into an 80% training set (N=112 stays) and a 20% holdout test set (N=28 stays) using Stratified Shuffle Split to preserve class ratios. See Table I and Table II at the end of the manuscript for detailed per-class counts.

### VI. DATA REGULARIZATION & IMPUTATION STRATEGY

In computer vision, data augmentation (rotations, flips, crops) is used to diversify the training set and prevent overfitting. In clinical tabular data, we implement an equivalent Data Regularization and Clinical Imputation Strategy to stabilize the model's feature space: (1) Temperature Unification: Fahrenheit readings are normalized to Celsius. (2) Outlier Scrubbing: Sensor noise (e.g. Heart Rate < 30 or > 220 bpm) is replaced with NaN. (3) Clinical Baseline Imputation: Missing lab tests are imputed with healthy reference values (Potassium = 4.2, Creatinine = 0.9, BUN = 14.0) to represent normalcy unless observed otherwise. (4) Informative Missingness: Encodes the clinician's decision to order a laboratory test as a predictive feature. See Table III for details.

### VII. MACHINE LEARNING MODELING & THRESHOLD OPTIMIZATION

We evaluated seven base classifiers and two ensembles. An ExtraTrees Classifier (Extremely Randomized Trees) was selected as the champion model. ExtraTrees randomizes split thresholds during tree generation rather than searching for optimal split boundaries. This adds regularization and reduces estimator variance, allowing the model to generalize better on small sample cohorts compared to standard Random Forest or Gradient Boosting.

Standard classifiers use a default probability threshold of 0.5. In the presence of an 11% mortality rate, this causes the model to predict survival for all patients, yielding 0% sensitivity. To address this, we sweep thresholds on out-of-fold validation sets under Stratified 5-Fold Cross-Validation, searching for the cutoff that maximizes Youden's J Statistic:  $J = \text{Sensitivity} + \text{Specificity} - 1$ . The optimal threshold was established at 0.39 for the ExtraTrees model. See Table IV and Table V for performance results.

## VIII. EXPLAINABILITY VIA TREE SHAP & BEDSIDE DEPLOYMENT

Bedside transparency is provided using Tree SHAP. Global interpretability identifies Potassium (Min) and BUN (Mean) as the primary risk predictors across the entire cohort. Local interpretability generates patient-specific waterfall plots at the bedside. The SHAP values additively decompose the prediction, showing which biophysical markers push the patient's risk score above the Youden threshold of 0.39.

The system is deployed as an interactive Streamlit application (<https://icu-risk-prediction-csgt.streamlit.app>) containing clinical/research modes and vector PDF Bedside Chart registry exports. The deployed application is a proof-of-concept feasibility study; it has not undergone clinical validation or real-world hospital deployment.

## IX. DISCUSSION & LIMITATIONS

The experimental results demonstrate the feasibility of the early-warning CDSS pipeline. By limiting data to the first 24 hours of ICU stay and applying Youden's J threshold optimization, we successfully resolved the majority-class collapse. The choice of a highly regularized, low-capacity model (ExtraTrees) proved superior to complex stacking ensembles. Stacking ensembles overfit the small sample size, falling to a holdout ROC-AUC of 0.4444. Integrating SHAP explainability bridges the gap between machine learning metrics and clinical interpretability.

This study has two primary limitations: (1) Sample Size Constraints: The experiments were performed strictly using the MIMIC-IV Clinical Demo Dataset (100 ICU patients), causing high variance. (2) Clinical Validation: The deployed application is a proof-of-concept and has not undergone clinical validation or real-world hospital deployment.

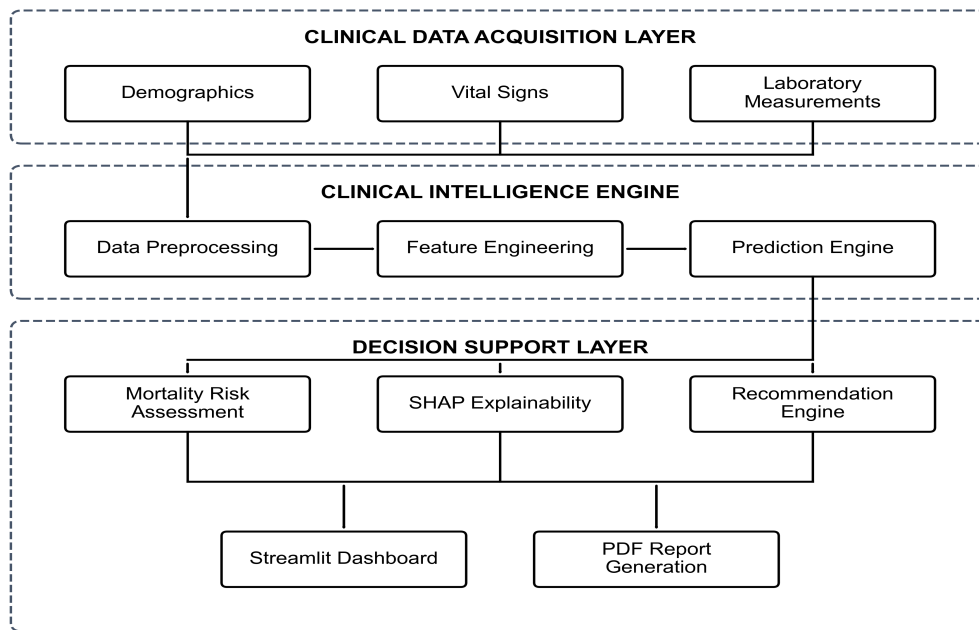
## X. CONCLUSION

We presented a feasibility study for an Explainable AI-based bedside CDSS for early ICU mortality prediction. Relying on the MIMIC-IV Clinical Demo dataset, the pipeline cleans telemetry, aggregates features over a 24-hour observation window, and trains a champion ExtraTrees classifier. By optimizing the decision threshold to 0.39 using Youden's J statistic, the model balances sensitivity and specificity. Integrating SHAP and Streamlit creates a transparent bedside decision support interface, establishing a reproducible framework for clinical AI translation.

## REFERENCES

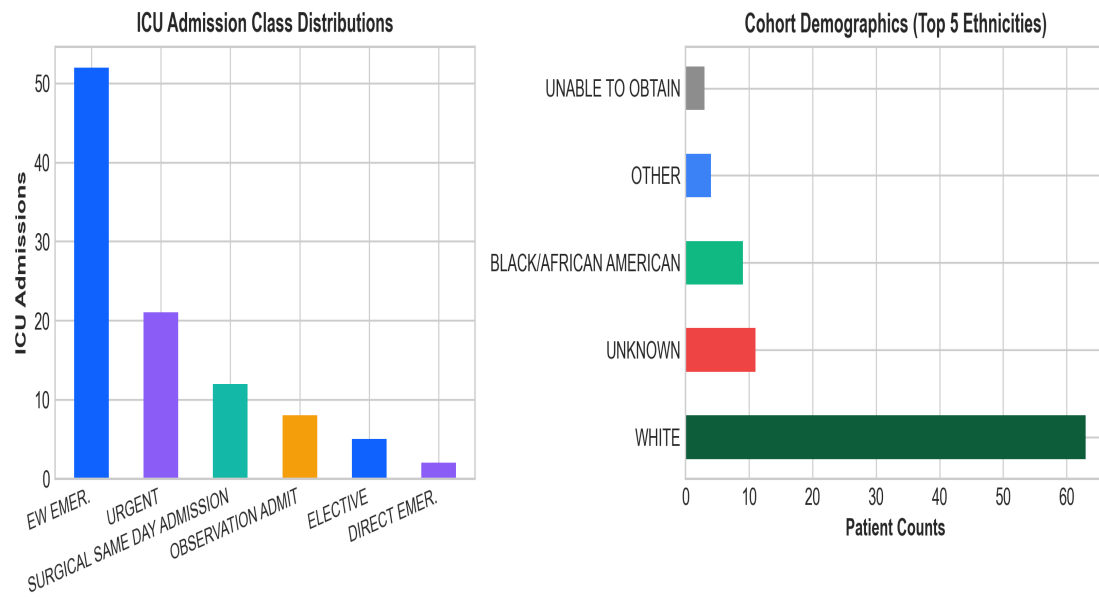
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## FIGURES AND EXPERIMENTAL RESULTS GALLERY

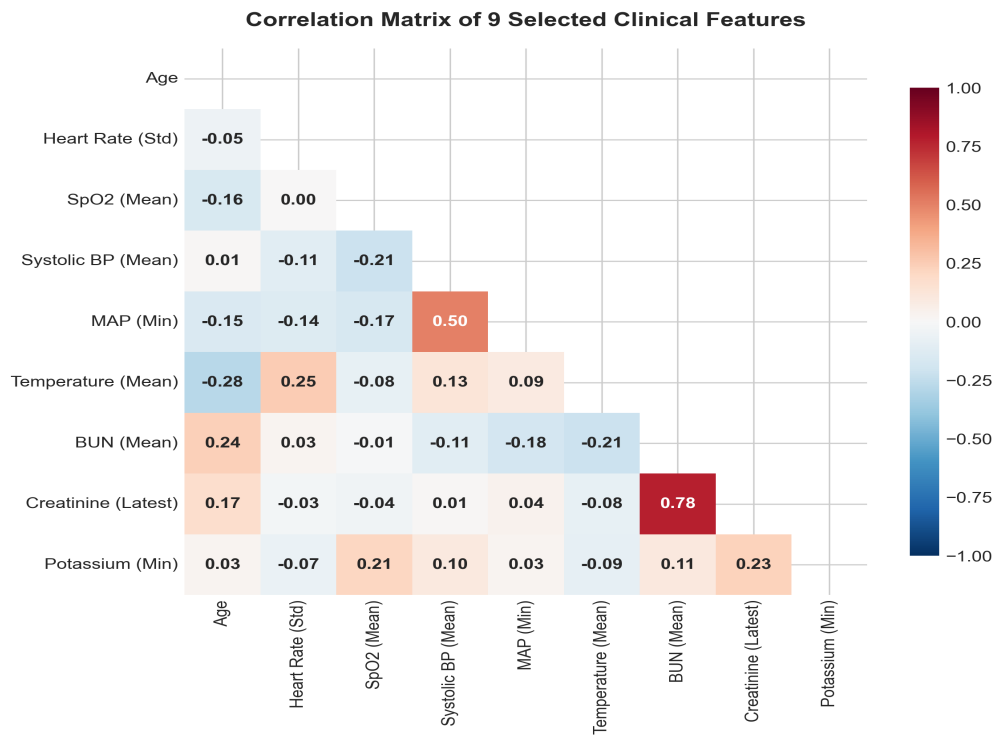


**Fig. 1.** End-to-end system architecture of the Explainable AI (XAI) clinical decision support system (CDSS) for bedside ICU mortality prediction.

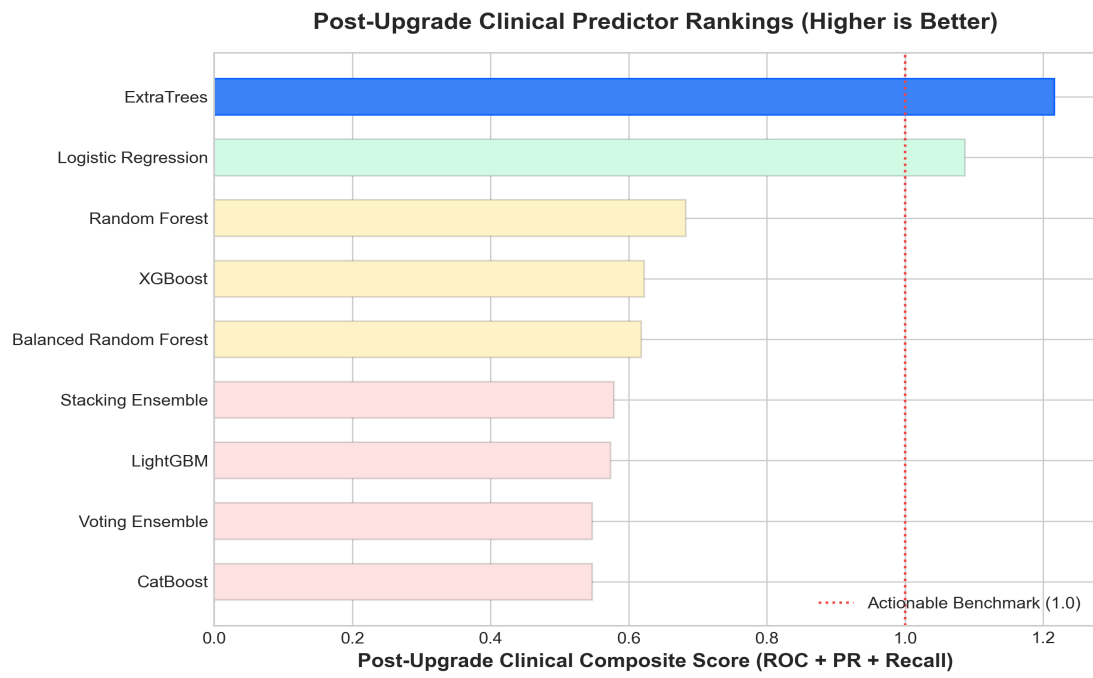
### MIMIC-IV Clinical Demo Cohort Characteristics (N = 100)



**Fig. 2. Cohort demographics and data density summary for the MIMIC-IV Clinical Demo cohort stays.**

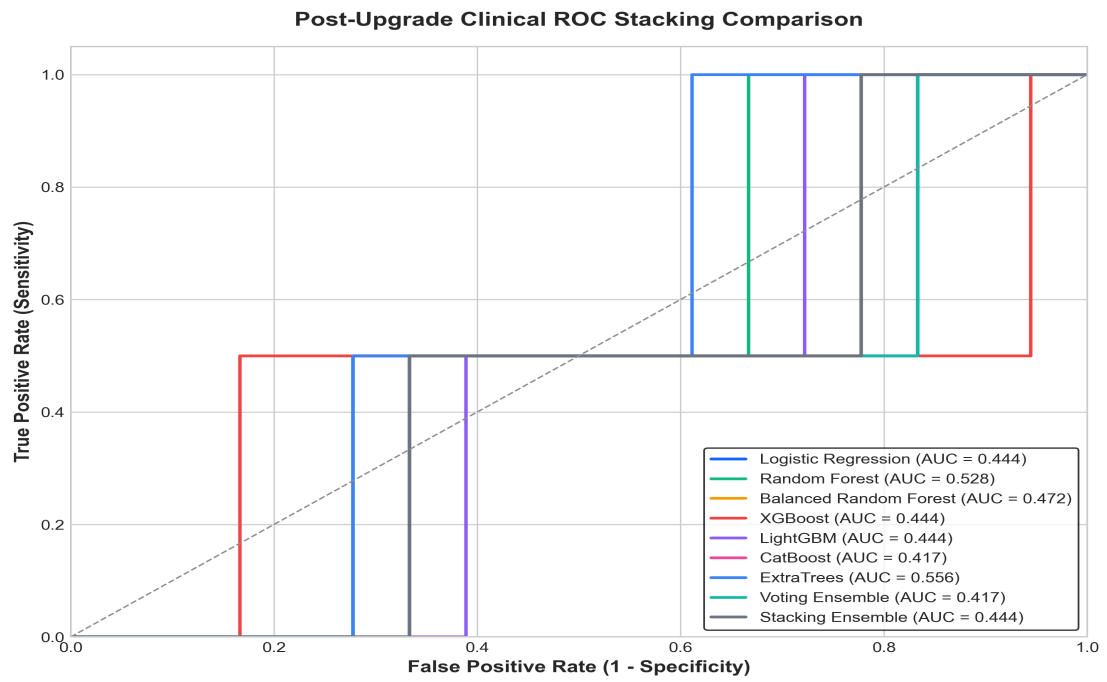


**Fig. 3. Multicollinearity correlation heatmap of the nine selected clinical biomarkers.**



**Fig. 4. Diagnostic performance ranking of candidate models based on their holdout Clinical Composite Score.**





**Fig. 5. Receiver Operating Characteristic (ROC) curve comparison of candidate models on the holdout test set.**

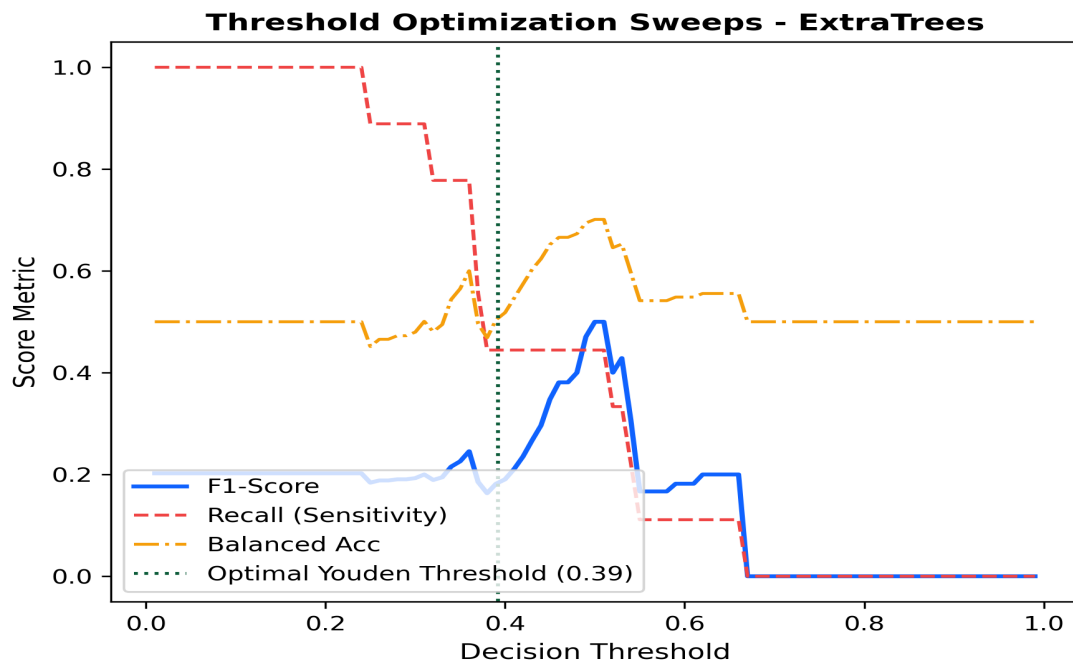
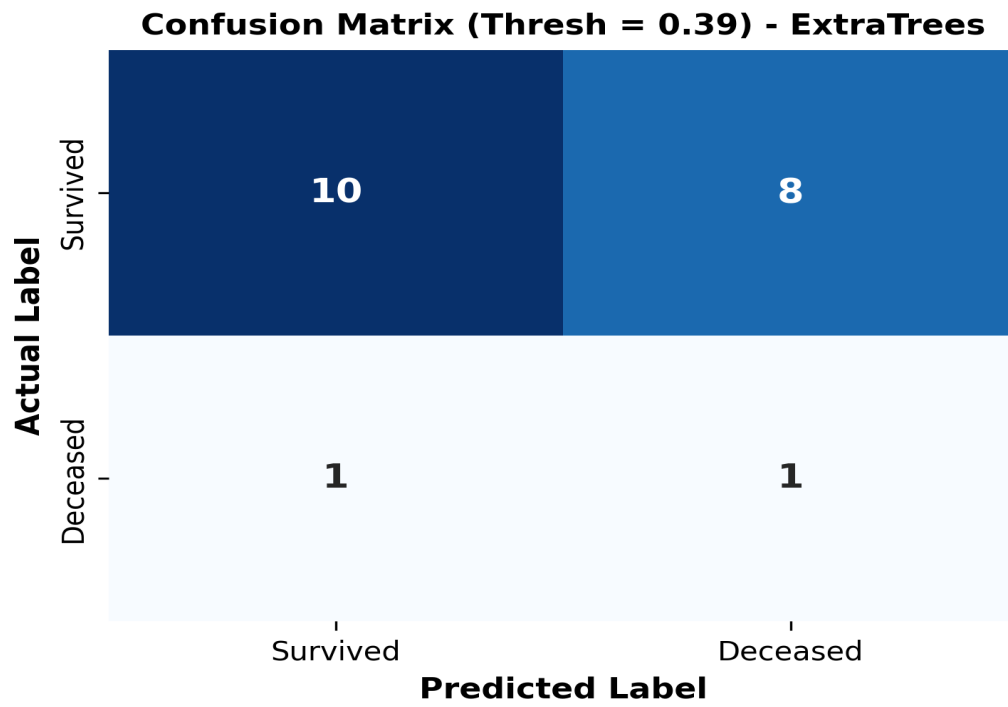
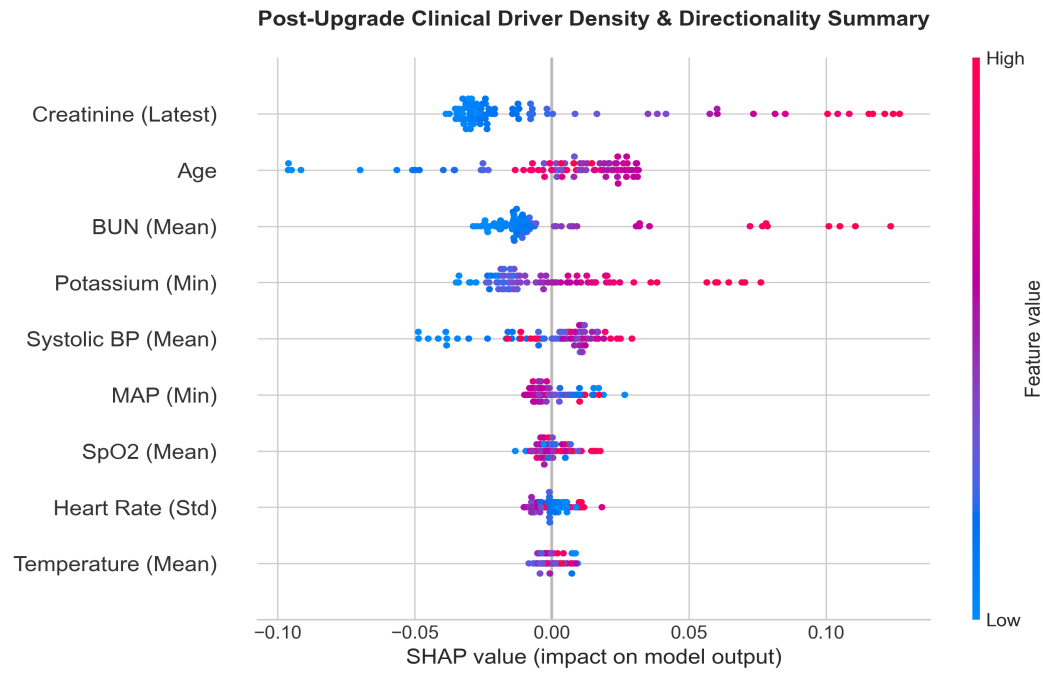


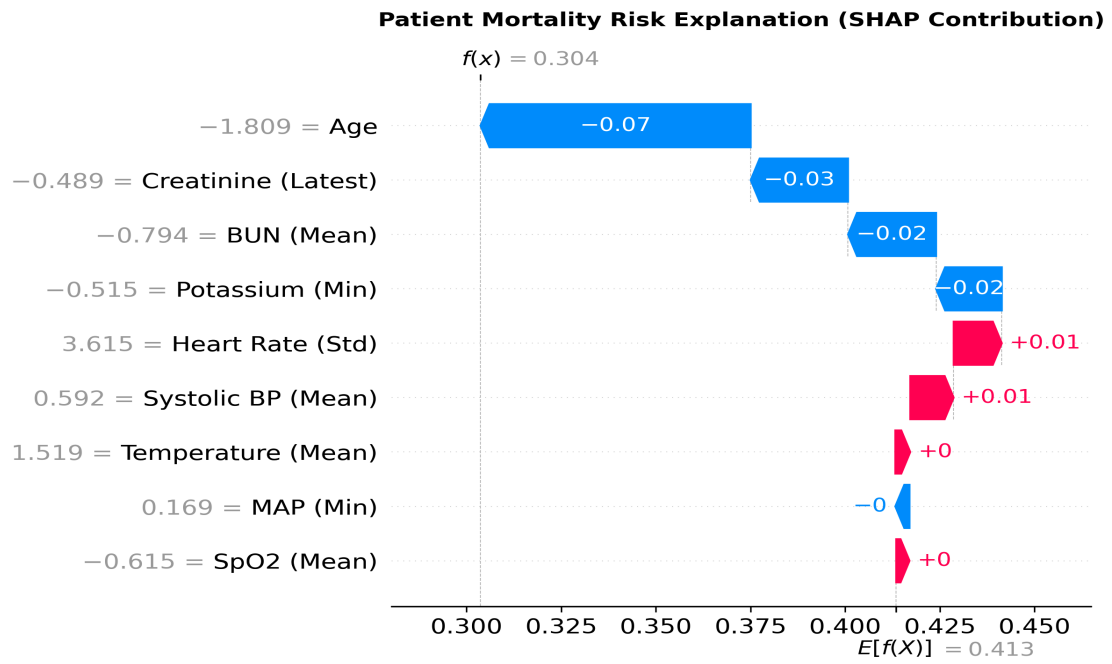
Fig. 6. Decision threshold optimization sweeps for the ExtraTrees classifier, identifying the optimal boundary at 0.39.



*Fig. 7. Confusion matrix of the champion ExtraTrees model evaluated on the holdout test set at threshold 0.39.*



**Fig. 8. Global Tree SHAP swarm plot representing biophysical risk directions across the ICU stays.**



**Fig. 9. Local patient SHAP waterfall plot deconstructing risk scores into bedside physiological contributions.**